15.10.26

Lemma 3.2 in $\mathbb{R}^{2\times 2}$

The two families of B to construct every normal N for any $A \in GL_2(\mathbb{R})$ in $\mathbb{R}^{2\times 2}$

When can we use lemma 3.2 of "Complex Matrix Inversion via Real Matrix Inversions", i.e. given a 2 by 2 real matrix A and after finding its QR decomposition, when is $N = Q^{-1}BR^{-1} = Q^TBR^{-1}$ normal? (given B of the form)

First of all, $A \in GL_2(\mathbb{R})$ so it has a unique QR decomposition.

for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} e & f \\ g & u \end{pmatrix}$, $\det(A) \neq 0 \Rightarrow ad - bc \neq 0$

$Q = \frac{1}{\sqrt{a^2+c^2}} \begin{pmatrix} a & -c \\ c & a \end{pmatrix}$ where $Q^T = Q^{-1}$ and $R = Q^TA = \frac{1}{\sqrt{a^2+c^2}} \begin{pmatrix} a^2+c^2 & ab+cd \\ 0 & -cb+ad \end{pmatrix}$

$C = A + iB = QR + iB = Q(R + iQ^TB) = Q(I + iQ^TBR^{-1})R$ and $N = Q^TBR^{-1}$

so $C = Q(I+iN)R$ but $\kappa(Q) = 1$ since Q is orthogonal and it preserves lengths → no stretching/shrinking

therefore, $\kappa(C) = \kappa((I+iN)\cdot R)$

for $s = \sqrt{a^2+c^2}$ and $M^{-1} = \begin{pmatrix} 1/x & -y/x^2 \\ 0 & 1/z \end{pmatrix}$ for $M = \begin{pmatrix} x & y \\ 0 & z \end{pmatrix}$: $R^{-1} = \begin{pmatrix} 1/s & t/-s\Delta \\ 0 & s/\Delta \end{pmatrix}$

where $t = ab+cd$ and $\Delta = ad-bc$

$$Q^T\cdot B = \frac{1}{s}\begin{pmatrix} a & c \\ -c & a \end{pmatrix}\begin{pmatrix} e & f \\ g & u \end{pmatrix} = \frac{1}{s}\begin{pmatrix} de+cg & af+cu \\ -ce+ag & -cf+au \end{pmatrix}$$

$$N = Q^TBR^{-1} = \frac{1}{s}\begin{pmatrix} de+cg & af+cu \\ -ce+ag & -cf+au \end{pmatrix}\begin{pmatrix} 1/s & t/-s\Delta \\ 0 & s/\Delta \end{pmatrix} = \begin{pmatrix} \frac{p}{s^2} & -\frac{pt}{s^2\Delta} + \frac{q}{\Delta} \\ \frac{r}{s^2} & -\frac{rt}{s^2\Delta} + \frac{u}{\Delta} \end{pmatrix}$$

where $p = ae+cg$, $q = af+cu$, $r = -ce+ag$, $u = -cf+au$

when is N normal? $NN^T = N^TN$, for $N = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$, $N^T = \begin{pmatrix} x & z \\ y & w \end{pmatrix}$

$$NN^T = \begin{pmatrix} x & y \\ z & w \end{pmatrix}\begin{pmatrix} x & z \\ y & w \end{pmatrix} = \begin{pmatrix} x^2+y^2 & xz+yw \\ xz+wy & z^2+w^2 \end{pmatrix}, \quad N^TN = \begin{pmatrix} x & z \\ y & w \end{pmatrix}\begin{pmatrix} x & y \\ z & w \end{pmatrix} = \begin{pmatrix} x^2+z^2 & xy+zw \\ xy+zw & y^2+w^2 \end{pmatrix}$$

- $x^2+y^2 = x^2+z^2 \Rightarrow y^2 = z^2 \Rightarrow y = \pm z$
- $xz+yw = xy+zw \Rightarrow xz-xy+yw-zw = 0 \Rightarrow x(z-y)-w(z-y) = 0 \Rightarrow (x-w)(y-z)$

So we have two cases:

case 1: $y = z$

case 2: $y = -z$ and $x = w$, one of the two cases must be true for N to be normal

$N = \begin{pmatrix} p/s^2 & -pt/s^2\Delta + q/\Delta \\ r/s^2 & -rt/s^2\Delta + u/\Delta \end{pmatrix}$ for case 1: $\frac{r}{s^2} = \frac{-pt}{s^2\Delta} + \frac{q}{\Delta} \Rightarrow r\Delta = -pt + s^2 q$

and for case 2: $\frac{r}{s^2} = \frac{pt}{s^2\Delta} - \frac{q}{\Delta} \Rightarrow r\Delta = pt - s^2 q$ and $\frac{p}{s^2} = \frac{-rt}{s^2\Delta} + \frac{u}{\Delta} \Rightarrow p\Delta = -rt + us^2$

case 1: $(ad+bc)(ag-ce) = -(ae+cg)(ab+cd) + (af+ch)(a^2+c^2)$

and after calculations we get: $af+ch = be+dg$

for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$ with $a^2+c^2 > 0$ (first row of A is nonzero) and $\det(A) \neq 0 \Rightarrow ad - bc \neq 0$ so that A^{-1} exists

we see that $A^TB = \begin{pmatrix} a & c \\ b & d \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+cg & af+ch \\ be+dg & bf+dh \end{pmatrix}$

and $(A^TB)^T = B^TA = \begin{pmatrix} ae+cg & be+dg \\ af+ch & bf+dh \end{pmatrix}$ so in case 1 N is normal if and only if A^TB is symmetric

even better: set $Y = BA^{-1} \Rightarrow B = YA$ thus $A^TB = A^TYA$ and so A^TB symmetric $\Leftrightarrow BA^{-1}$ symmetric

that's because A^TYA must be symmetric: $(A^TB)^T = A^TB \Rightarrow (A^TYA)^T = A^TYA$

$\Rightarrow A^TY^TA = A^TYA \Rightarrow A^T(Y^T - Y)A = 0 \Rightarrow Y^T = Y$ since A is invertible so $A \neq 0$

we have shown that A^TB symmetric $\Leftrightarrow BA^{-1}$ symmetric $\Leftrightarrow$ N is normal in case 1 so $B = YA$ and can be obtained by multiplying A by a symmetric matrix on the left, this is a way to generate for our A

results for case 1: for any fixed invertible matrix $A \in \mathbb{R}^{2\times 2}$, every matrix B, satisfying case 1, has the form $B = \begin{pmatrix} p & q \\ q & r \end{pmatrix}A$, $p,q,r \in \mathbb{R}$

this can also be derived from condition $y = z \Rightarrow N = \begin{pmatrix} x & y \\ y & w \end{pmatrix}$: symmetric

since $B = SA$, for S: symmetric $\Rightarrow B = SQR$ and $N = Q^TSQRR^{-1} = Q^TSQ$,

$N^T = Q^TS^TQ$ but $S = S^T \Rightarrow N^T = Q^TSQ = N$ so N is also symmetric

Because S can be any symmetric 2 by 2 matrix and A can be any invertible matrix, this construction can produce every symmetric N which is also normal

$C = A + iB = Q(I + iS)R$

case 2: $N=\begin{pmatrix} x & y \\ -y & x \end{pmatrix} = xI+yJ$ for $J=\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $J^2=-I$

$B=QNR=Q(xI+yJ)R=(xI+yQJQ^T)A$

so for any invertible A we can choose any real x,y so that we construct B for N to be normal with $B=(xI+yQJQ^T)A$, like $B=SA$ in case 1

in case 2 we construct a family of matrices that are not symmetric.

The two cases produce every normal matrix in $\mathbb{R}^{2\times 2}$ because N is normal it is of the form $N=\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ or $N=\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$

This is a standard result in linear algebra.

Therefore, we have proved that all normal matrices $N\in\mathbb{R}^{2\times 2}$ are attainable for any fixed invertible A by setting $B=SA$ or $B=(xI+yQJQ^T)A$.

In case 1: $C_1=A+iB=A+iSA=(I+iS)A$.

In case 2: $C_2=Q(I+iN)R=Q\left[I+i\begin{pmatrix} x & y \\ -y & x \end{pmatrix}\right]R$

I could investigate how $\kappa(C)$ is affected in both cases by S's entries in the first one and x,y in the second for a fixed A. Could start with $A=\begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}$ to test $\kappa(A)$ and compare it to $\kappa(C)$ and then all diagonals $A=\begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}$.

We care for N being normal since that we can use that normal matrices are unitarily diagonizable: $N=U\Lambda U^H$, where $UU^H=U^HU=I$ and $\Lambda=\mathrm{diag}(\lambda_1,\dots,\lambda_n)$

proof: for λ eigenvalue of N, $Nv=\lambda v \Rightarrow U\Lambda U^*v=\lambda v \Rightarrow U^*U\Lambda U^*v=\lambda U^*v$

$\Rightarrow \Lambda U^*v=\lambda U^*v$, $w=U^*v$, $\Lambda w=\lambda w$, $w\neq 0$ since $U\in GL_n(\mathbb{C})$ and $v\neq 0$

therefore, eigenvalues of N are the eigenvalues of Λ.

ex: for $A=\begin{pmatrix} 1 & 0 \\ 0 & 10 \end{pmatrix}$, $S=\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, $Q=I$, $R=A$, $B=QSR=\begin{pmatrix} s_1 & 0 \\ 0 & as_2 \end{pmatrix}=\begin{pmatrix} 2 & 0 \\ 0 & 10 \end{pmatrix}$, case 1

$C=(I+iS)A=\begin{pmatrix} 1+i2 & 0 \\ 0 & 10(1+i) \end{pmatrix}$, $\kappa(A)=10$, $\kappa(C)=2\sqrt{10}\simeq 6.325$ so $\kappa(A)>\kappa(C)$

Prove that for $A=\begin{pmatrix}1&0\\0&\epsilon\end{pmatrix}$, $A=QR$ and $B=(xI+yQJQ^T)A$ for N to be normal, one of the two cases, $B=QNR \Rightarrow N=Q^TBR^{-1}$ we cannot improve the conditioning of $C=A+iB=Q(I+iN)R$, so $\kappa_2(C)\geq\kappa_2(A)$.

say $0<\epsilon\ll 1$ $N=\begin{pmatrix}x&y\\-y&x\end{pmatrix}$, $B=QNR=NA=\begin{pmatrix}x&\epsilon y\\-y&\epsilon x\end{pmatrix}$ for $Q=I, R=A$

$C=A+iB=\begin{pmatrix}1&0\\0&\epsilon\end{pmatrix}+i\begin{pmatrix}x&\epsilon y\\-y&\epsilon x\end{pmatrix}=\begin{pmatrix}1+ix&i\epsilon y\\-iy&\epsilon(1+ix)\end{pmatrix}$

so $C^*C=\begin{pmatrix}1+x^2+y^2&2i\epsilon y\\-2i\epsilon y&\epsilon^2(1+x^2+y^2)\end{pmatrix}$, $\det(C^*C)=0 \Rightarrow \lambda_\pm=\frac{s(1+\epsilon^2)\pm\sqrt{s^2(1-\epsilon^2)^2+16\epsilon^2y^2}}{2}$

$\kappa_2(C)=\sqrt{\frac{\lambda_+}{\lambda_-}}$ for $s=1+x^2+y^2$

we know that $\kappa_2(C)\geq\frac{\max(\|C_1\|_2,\|C_2\|_2)}{\min(\|C_1\|_2,\|C_2\|_2)}$, $C=\begin{pmatrix}1+ix&i\epsilon y\\-iy&\epsilon(1+ix)\end{pmatrix}$,

where $\|C_1\|_2=\sqrt{1+x^2+y^2}$, $\|C_2\|_2=\sqrt{\epsilon^2(1+x^2+y^2)}=\epsilon\sqrt{1+x^2+y^2}=\epsilon\|C_1\|_2$

with $0<\epsilon\ll 1$ so $\|C_1\|_2>\|C_2\|_2$ and $\kappa_2(C)\geq\frac{\|C_1\|_2}{\|C_2\|_2}$

therefore, $\kappa_2(C)\geq\frac{1}{\epsilon}$ but $\kappa_2(A)=\frac{1}{\epsilon}$ so $\kappa_2(C)\geq\kappa_2(A)$. □

- Prove that $\kappa_2(C)\geq\frac{\|C_1\|_2}{\|C_2\|_2}$, $C=\begin{pmatrix}|&|\\C_1&C_2\\|&|\end{pmatrix}$ with $\|C_1\|_2\geq\|C_2\|_2$.

$\kappa_2(C)=\|C\|_2\cdot\|C^{-1}\|_2$, by definition $\|A\|_2=\max_{x\neq0}\frac{\|Ax\|_2}{\|x\|_2}\geq\frac{\|Ax\|_2}{\|x\|_2}$

so $\|C\|_2\geq\frac{\|Ce_1\|_2}{\|e_1\|_2}$ but $Ce_1=C_1$ and $\|e_1\|_2=1 \Rightarrow \|C\|_2\geq\|C_1\|_2$.

$Ce_2=C_2 \Rightarrow C^{-1}C_2=e_2$: $\|C^{-1}\|_2\geq\frac{\|C^{-1}C_2\|_2}{\|C_2\|_2} \Rightarrow \|C^{-1}\|_2\geq\frac{1}{\|C_2\|_2}$

$\|C_1\|_2\cdot\|C^{-1}\|_2\geq\frac{\|C_1\|_2}{\|C_2\|_2} \Rightarrow \kappa_2(C)\geq\frac{\|C_1\|_2}{\|C_2\|_2}$. □

- Prove the same for any $A=\mathrm{diag}(a,b)$, $a,b\neq0$, $N=\begin{pmatrix}x&y\\-y&x\end{pmatrix}$, $A=QR$, $Q=I$, $R=A$.

$B=NA=\begin{pmatrix}xa&yb\\-ya&xb\end{pmatrix}$, $C=\begin{pmatrix}a(1+ix)&iyb\\-iya&b(1+ix)\end{pmatrix}$, $C_1=\begin{pmatrix}a(1+ix)\\-iya\end{pmatrix}$, $C_2=\begin{pmatrix}iyb\\b(1+ix)\end{pmatrix}$

$\|C_1\|_2=a\sqrt{1+x^2+y^2}$, $\|C_2\|_2=b\sqrt{1+x^2+y^2}$, $\kappa_2(C)\geq\frac{\|C_1\|_2}{\|C_2\|_2} \Rightarrow \kappa_2(C)\geq\frac{\max\{a,b\}}{\min\{a,b\}}$

but $\kappa_2(A)=\frac{\max\{a,b\}}{\min\{a,b\}}$, so $\kappa_2(C)\geq\kappa_2(A)$. □

· Prove the above for any $A=QR$.

$A=\begin{pmatrix}a&0\\0&b\end{pmatrix}$, $N=\begin{pmatrix}x&y\\-y&x\end{pmatrix}$, $B=QNR$, $C=A+iB$, $R=Q^TA$, $C=(I+iQNQ^T)A$

we know that $Q^*JQ^T=\det(Q)J$ since for $Q=\begin{pmatrix}p&r\\q&s\end{pmatrix}$

$QJQ^T=\begin{pmatrix}-pq+pq & -ar+ps\\ -sp+qr & -sr+rs\end{pmatrix}=\begin{pmatrix}0 & ps-qr\\ qr-ps & 0\end{pmatrix}=\det(Q)J$, since $\det(Q)=\pm1$

$QNQ^T=xI+y\det(Q)J$, set $y\det(Q)=t\in\mathbb{R}$

then $C=(I+i(xI+tJ))A$, $C=\begin{pmatrix}a(1+ix) & ib\\ -ia & b(1+ix)\end{pmatrix}$

same as the last proof so in this case $\kappa_2(C)\geq\kappa_2(A)$ holds too. □

· for $A=\begin{pmatrix}a&0\\0&b\end{pmatrix}$, $N=\begin{pmatrix}p&q\\q&r\end{pmatrix}$, $B=QNR$, try to make a perfectly conditioned C

Say that $0<a<b$ since we don't lose generality and we could use $\kappa_2(A)=\frac{\max(|a|,|b|)}{\min(|a|,|b|)}$ but it would mean more calculations.

$\kappa_2(A)=\frac{b}{a}$, $B=SA$ for $S=QNQ^T$ because $R=Q^TA$ and $B=QNQ^TA=SA$

N: symmetric $\Rightarrow$ S: symmetric as we have shown.

$B=\overbrace{\begin{pmatrix}u&v\\v&w\end{pmatrix}}^{S}\overbrace{\begin{pmatrix}a&0\\0&b\end{pmatrix}}^{A}=\begin{pmatrix}ua&vb\\va&wb\end{pmatrix}$, $C=A+iB=\begin{pmatrix}a(1+iu)&ivb\\iva&b(1+iw)\end{pmatrix}$

Can some u,v,w give us $\kappa_2(C)$ always?

we must prove that $\kappa_2(C)=1\iff C^*C=\lambda I$ for some $\lambda>0$

proof: $\kappa_2(C)=\frac{\sigma_{max}(C)}{\sigma_{min}(C)}$, $\kappa_2(C)=1\iff\sigma_{max}(C)=\sigma_{min}(C)$

so all singular values of C must be equal let's say to s

$(C^*C)^*=C^*C$ so C: Hermitian and so diagonizable by a unitary matrix U

$C^*C=U\begin{pmatrix}s^2&&0\\&\ddots&\\0&&s^2\end{pmatrix}U^*$ so $C^*C=UsIU^*=s^2UU^*=s^2I\Rightarrow C^*C=\lambda I$

for $C^*C=\lambda I\Rightarrow\kappa_2(C)=\frac{\sqrt{\lambda}}{\sqrt{\lambda}}=1$

so $C^*C=\begin{pmatrix}a^2(1+x^2+y^2) & aby(x+z)\\ aby(x+z) & b^2(1+z^2+y^2)\end{pmatrix}=\lambda I$ for $\kappa_2(C)=1$

so $aby(x+z)=0\xrightarrow{ab\neq0}y(x+z)=0$, for $x+z=0\Rightarrow x=-z$ (impossible: $a<b$)

$a^2(1+x^2+y^2)=b^2(1+z^2+y^2)\Rightarrow a^2(1+z^2+y^2)=b^2(1+z^2+y^2)\Rightarrow a^2=b^2$

so $y=0$ and so $N=\begin{pmatrix}x&0\\0&z\end{pmatrix}$

and solving $a^2(1+x^2+y^2)=b^2(1+z^2)$ we get that all symmetric matrices of this family $N=\begin{pmatrix}x&y\\y&z\end{pmatrix}$ that achieve perfect conditioning for C are of the form: $N=\begin{pmatrix}\pm\sqrt{\frac{b^2}{a^2}(1+z^2)-1}&0\\0&z\end{pmatrix}$, $z\in\mathbb{R}$

So for all matrices of this family: $B=QNR$, $A=QR$, $C=A+iB$ we can achieve perfect conditioning for C, $\kappa_2(C)$, by choosing $N=\begin{pmatrix} u & 0 \\ 0 & w \end{pmatrix}$ with $\left(\frac{a}{b}\right)^4 < \frac{1+w^2}{1+u^2} < 1$ for $\kappa_2(C) < \kappa_2(A)$ where $A=\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$. □

- Let's prove the same for every $A \in GL_2(\mathbb{R})$.

$A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det A \neq 0$, $A=QR$, $R=\begin{pmatrix} r & s \\ 0 & t \end{pmatrix}$, $rt \neq 0$, $Q^TQ=I$ since $Q \in O_{n2}(\mathbb{R})$

$B=QNR$, $C=Q(I+iN)R$, $\kappa_2(C)=\kappa_2((I+iN)R)$ and $\kappa_2(A)=\kappa_2(R)$

we know that we could try $N=\begin{pmatrix} x & y \\ y & z \end{pmatrix}$, $x,y,z \in \mathbb{R}$, since it has worked for diagonal A, $I+iN=\begin{pmatrix} 1+ix & iy \\ iy & 1+iz \end{pmatrix}$

$M=(I+iN)R=\begin{pmatrix} r(1+ix) & s(1+ix)+ity \\ iry & isy+t(1+iz) \end{pmatrix}$ we want $\kappa_2(M)=1 \Rightarrow M^*M=\lambda I$, $\lambda>0$

for $M=\begin{pmatrix} p & q \\ u & v \end{pmatrix}$, $M^*=\begin{pmatrix} \bar{p} & \bar{u} \\ \bar{q} & \bar{v} \end{pmatrix}$, $a\cdot\bar{a}=|a|^2$, $M^*M=\begin{pmatrix} |p|^2+|u|^2 & \bar{p}q+\bar{u}v \\ \bar{q}p+\bar{v}u & |q|^2+|v|^2 \end{pmatrix}=\lambda I$

$$\begin{cases} \bar{p}q+\bar{u}v=0 \\ \bar{q}p+\bar{v}u=0 \\ |p|^2+|u|^2=|q|^2+|v|^2 \end{cases}$$

$p=r(1+ix)$, $q=s(1+ix)+ity$, $u=iry$, $v=isy+t(1+iz)$

$(M^*M)_{12}=0 \Rightarrow r[s(1+x^2+y^2)+ty(x+z)]=0 \overset{r\neq 0}{\Rightarrow}$

$s(1+x^2+y^2)+ty(x+z)=0$ (1)

$(M^*M)_{11}=r^2(1+x^2+y^2)$, $(M^*M)_{22}=|q|^2+|v|^2=s^2(1+x^2+y^2)+t^2(1+y^2+z^2)+2sty(x+z)$

so $r^2(1+x^2+y^2)=s^2(1+x^2+y^2)+t^2(1+y^2+z^2)+2sty(x+z)$ (2)

from (1): $2sty(x+z)=-2s^2(1+x^2+y^2)$

(2) $\overset{(1)}{\Rightarrow}$ $r^2(1+x^2+y^2)=s^2(1+x^2+y^2)+t^2(1+y^2+z^2)-2s^2(1+x^2+y^2)$

$\Rightarrow (r^2+s^2)(1+x^2+y^2)=t^2(1+y^2+z^2)$

$(M^*M)_{12}=\overline{\bar{p}q+\bar{u}v}=\bar{q}p+\bar{v}u=(M^*M)_{21}$ so they give us the same equation

Therefore, $\begin{cases} s(1+x^2+y^2) + ty(x+z) = 0 \quad (1) \\ (r^2+s^2)(1+x^2+y^2) = t^2(1+y^2+z^2) \quad (2) \end{cases}$ $R = \begin{pmatrix} r & s \\ 0 & t \end{pmatrix}$, $N = \begin{pmatrix} x & y \\ y & z \end{pmatrix}$, $M = (I+iN)R$

we can choose $x, y, z \in \mathbb{R}$ to help us with calculations, let's try $x = -\frac{t}{r}$, $y = \frac{s}{r}$

$1+x^2+y^2 = \frac{r^2+t^2+s^2}{r^2}$

$s(1+x^2+y^2) + ty(x+z) = 0 \Rightarrow s\frac{r^2+t^2+s^2}{r^2} + \frac{ts}{r}\left(-\frac{t}{r}+z\right) = 0 \Rightarrow$

$s\frac{r^2+s^2}{r^2} + \frac{st^2}{r^2} - \frac{st^2}{r^2} + z\frac{ts}{r} = 0 \Rightarrow s\frac{r^2+s^2}{r^2} + z\frac{ts}{r} = 0$, multiply by $\frac{r^2}{st}$, $st \neq 0$

$\Rightarrow r^2+s^2+rtz = 0 \Rightarrow z = -\frac{r^2+s^2}{rt}$, if $s = 0$, (1) is satisfied and $z = -\frac{r}{t}$, $y = 0$

$(2) \Rightarrow (r^2+s^2)(1+x^2+y^2) = t^2(1+y^2+z^2) \Rightarrow \frac{(r^2+s^2)(r^2+s^2+t^2)}{r^2} = t^2\left(1+\frac{s^2}{r^2}+\frac{(r^2+s^2)^2}{r^2t^2}\right)$

RHS $t^2\left(1+\frac{s^2}{r^2}+\frac{(r^2+s^2)^2}{r^2t^2}\right) = t^2\left(\frac{r^2t^2+s^2t^2+(r^2+s^2)^2}{r^2t^2}\right) = \frac{t^2(r^2+s^2)+(r^2+s^2)^2}{r^2}$

$= \frac{(t^2+r^2+s^2)(r^2+s^2)}{r^2} =$ LHS so LHS = RHS and the equations are satisfied for $x = -\frac{t}{r}$, $y = \frac{s}{r}$, $z = -\frac{r^2+s^2}{rt}$

$N = \begin{pmatrix} x & y \\ y & z \end{pmatrix} = \begin{pmatrix} -t/r & s/r \\ s/r & -\frac{r^2+s^2}{rt} \end{pmatrix}$, $B = QNR$, $C = Q(I+iN)R$, $R = \begin{pmatrix} r & s \\ 0 & t \end{pmatrix}$

$I+iN = \begin{pmatrix} 1-i\frac{t}{r} & i\frac{s}{r} \\ i\frac{s}{r} & 1-i\frac{r^2+s^2}{rt} \end{pmatrix}$, $M = (I+iN)R = \begin{pmatrix} r-it & s+i\frac{st}{r} \\ is & t-i\frac{r^2+s^2}{r} \end{pmatrix}$

remember that $\kappa_2(C) = \kappa_2(M)$, we proved that for these choices of x, y, z, M^*M has the form $\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \lambda I \Rightarrow M^*M = \lambda I$

but $(M^*M)_{11} = r^2(1+x^2+y^2) = r^2+s^2+t^2$, so $\lambda = r^2+s^2+t^2$

and $M^*M = (r^2+s^2+t^2)I$

also $\|Mv\|_2^2 = v^*M^*Mv = \lambda\|v_2\|^2$ so M scales every vector by $\sqrt{\lambda}$

$= \sqrt{r^2+s^2+t^2}$, $\sigma_{MAX}(M) = \sigma_{MIN}(M) = \sqrt{\lambda}$ so $\kappa_2(M) = \frac{\sqrt{\lambda}}{\sqrt{\lambda}} = 1 = \kappa_2(C)$. □

How to make C perfectly conditioned for any $A \in GL_n(\mathbb{R})$

• For $A \in GL_n(\mathbb{R})$ find B for $C = A + iB$ where $\kappa_2(C) = 1$. Prove that we can make $C = A + iB$ perfectly conditioned for any invertible A in $\mathbb{R}^{n \times n}$.

$C = Q(I+iN)R$ where $A = QR$, $B = QNR$ and we know that:

$\kappa_2(C) = 1 \iff C^*C = \lambda I$, $\lambda > 0$

$C^* = \overline{(Q(I+iN)R)}^T = R^T \overline{(I+iN)}^T Q^T$ where $(I+iN)^T = I - iN$ since $N^T = N$ for N normal so $C^* = R^T(I - iN)Q^T$

and $C^*C = R^T(I-iN)Q^T Q(I+iN)R = R^T(I-iN)(I+iN)R$

but $(I-iN)(I+iN) = I + iN - iN - iN \cdot N = I + N^2$

and $C^*C = R^T(I+N^2)R = \lambda I \Rightarrow N^2 = \lambda R^{-T}R^{-1} - I = \lambda(RR^T)^{-1} - I$

we set $M = N^2 = \lambda R^{-T}R^{-1} - I$ and M is positive semidefinite

because $M = N^2 \Rightarrow N = M^{1/2}$

because $R^{-T}R^{-1} = (R^{-1})^T R^{-1}$ and $x^T R^{-T}R^{-1}x = (R^{-1}x)^T(R^{-1}x) = \|R^{-1}x\|_2^2 \geq 0$

so $R^{-T}R^{-1}$ is positive definite, for $\mu_1, \dots, \mu_n$ the eigenvalues of $R^{-T}R^{-1}$ we want M to be positive semidefinite then the eigenvalues of $\lambda R^{-T}R^{-1} - I$ are $\lambda\mu_1 - 1, \dots, \lambda\mu_n - 1$ and $\lambda\mu_{min} - 1 \geq 0 \Rightarrow \boxed{\lambda \geq \frac{1}{\mu_{min}}}$

also M is symmetric: $M = \lambda R^{-T}R^{-1} - I$, $M^T = \lambda(R^{-T}R^{-1})^T - I = \lambda R^{-T}R^{-1} - I = M$

so $M^T = M$ and M is therefore symmetric positive semidefinite matrix and for every real symmetric matrix can be diagonalized by an orthogonal matrix: $\exists U \in O_n(\mathbb{R})$ and Λ diagonal, such that $M = U\Lambda U^T$

for $M = U\Lambda U^T$ and $\Lambda = diag(\lambda_1, \lambda_2, \dots, \lambda_n)$, $D = C(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \dots, \sqrt{\lambda_n})$

so $D^2 = \Lambda \Rightarrow U = \Lambda^{1/2}$

try $N = UDU^T$ to solve $N^2 = M$, $N = UDU^T = U\Lambda^{1/2}U^T$

and $N^2 = (UDU^T)(UDU^T) = UD(U^TU)DU^T = UD^2U^T = U\Lambda U^T = M$

so $N = UDU^T$ is a solution for $N^2 = M$ where $M = U\Lambda U^T$

so $I + N^2 = \lambda R^{-T}R^{-1}$ and $C^*C = R^T(\lambda R^{-T}R^{-1})R = R^T\lambda R^{-T} \Rightarrow C^*C = \lambda I$

as we assumed and $\sigma_1(C) = \sigma_2(C) = \dots = \sigma_n(C) = \sqrt{\lambda}$

Therefore, $\kappa_2(C) = \frac{\sigma_{max}(C)}{\sigma_{min}(C)} = \frac{\sqrt{\lambda}}{\sqrt{\lambda}} = 1 \rightarrow C$ is perfectly conditioned

for $N = U\Lambda^{1/2}U^T$ and $N^2 = M = U\Lambda U^T$

we must choose $\lambda \geq \frac{1}{\lambda_{min}(R^{-T}R^{-1})}$ and $N = (\lambda R^{-T}R^{-1} - I)^{1/2} \Rightarrow N^2 = \lambda R^{-T}R^{-1} - I$

where $B = QNR$ and $C = A + iB$ $\quad \lambda \geq \frac{1}{\mu_{min}} \rightarrow R^{-T}R^{-1}$

The point is to make $N^2 = M$ positive semidefinite for big enough $\lambda > 0$. □

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- Given $C=A+iB$, we have B such that $\kappa_2(A+iB)=\kappa_2(C)=1 \quad \forall A\in GL_n(\mathbb{R})$. This is given from $N^2=M=aR^{-T}R^{-1}-I$ with $a\geq \sigma_{max}(R)=\sigma_{max}(A)$ for $A=QR$ and $B=QNR$. How costly is this construction?

For $R^{-T}R^{-1}=(RR^T)^{-1}$, then $M=a(RR^T)^{-1}-I=N^2$
the eigenvalues of RR^T are $\sigma(R)^2,\dots,\sigma_n(R)^2$ since $\lambda_i(R^TR)=\sigma_i(R)^2$
so the eigenvalues of $(RR^T)^{-1}$ are $\frac{1}{\sigma_1(R)^2},\dots,\frac{1}{\sigma_n(R)^2}$
and the eigenvalues of M are $\lambda_i(M)=\frac{a}{\sigma_i(R)^2}-1$.
We want M to be positive semidefinite, $M\succeq 0$: $\lambda_i(M)\geq 0 \Rightarrow a\geq \sigma_i(R)^2$

$\sigma_{max}(R)=\|R\|_2$, so $a\geq\|R\|_2^2$, $B=QNR$ so $\|B\|_2=\|NR\|_2$, for $\|Q\|_2=1$

$\|B\|_2^2=\|B^TB\|_2$, $B^TB=(QNR)^T(QNR)=R^TN^TQ^TQNR \overset{Q^T=Q^{-1}}{=} R^TN^TNR$
$=R^TN^2R=R^TMR$, because N is symmetric and for $M=N^2$

but $M=aR^{-T}R^{-1}-I$ so $B^TB=R^T(aR^{-T}R^{-1}-I)R=aI-R^TR$
but the eigenvalues of R^TR are $\sigma_i(R)^2$ so the eigenvalues of B^TB are $a-\sigma_i(R)^2$.

$\|B\|_2^2=\max_i(a-\sigma_i(R)^2)$, since $\|B\|_2=\sqrt{\lambda_{max}(B^TB)}$, with $a\geq\sigma_i(R)^2$

So $\|B\|_2^2=\lambda_{max}(B^TB)=a-\sigma_{min}(R)^2 \Rightarrow \|B\|_2=\sqrt{a-\sigma_{min}(R)^2}$
Q is orthogonal and A and R have the same singular values since $A^TA=R^TR$, therefore,

$\|B\|_2=\sqrt{a-\sigma_{min}(A)^2}$, for $a\geq\sigma_{max}(A)^2$ we have that $a_{min}=\sigma_{max}(A)^2$

In this case $\|B\|_{2min}=\sqrt{\sigma_{max}(A)^2-\sigma_{min}(A)^2}$.

For the construction above and for the smallest allowed a, we have an exact formula for the size of B, not just an upper bound.

$$\kappa_2(A) = \frac{\sigma_{max}(A)}{\sigma_{min}(A)} \Rightarrow \sigma_{min}(A) = \frac{\sigma_{max}(A)}{\kappa_2(A)}$$

$$\|B\|_2 = \sqrt{\sigma_{max}(A)^2 - \frac{\sigma_{max}(A)^2}{\kappa_2(A)^2}} = \sigma_{max}\sqrt{1 - \frac{1}{\kappa_2(A)}}$$

Therefore, as $\kappa_2(A) \to \infty$, so A is arbitrarily ill-conditioned, we get that $\|B\|_2 \to \sigma_{max}(A)$, provided that $\sigma_{max}(A)$ is fixed.

To sum up, $\|B\|_2 = \sqrt{a - \sigma_{min}(A)^2}$ and $\|B\|_{2\,min} = \sqrt{\sigma_{max}(A)^2 - \sigma_{min}(A)^2}$

$\kappa_2(B) = \frac{\sigma_{max}(A)}{\sigma_{min}(A)}$, but $\sigma_{max}(B) = \sqrt{a - \sigma_{min}(A)^2}$ and $\sigma_{min}(B) = \sqrt{a - \sigma_{max}(A)^2}$

so $\kappa_2(B) = \sqrt{\frac{a - \sigma_{min}(A)^2}{a - \sigma_{max}(A)^2}}$ where $a \geq \sigma_{max}(A)$ for our construction to result to C being perfectly conditioned, $\kappa_2(C) = 1$. □

- Can we control $\|B\|_2$ and $\kappa_2(B)$ simultaneously?

$\|B\|_2 = \sqrt{a - \sigma_{min}(A)^2}$, $\kappa_2(B) = \|B\|_2 \cdot \|B^{-1}\|_2 = \sqrt{\frac{a - \sigma_{min}(A)^2}{a - \sigma_{max}(A)^2}}$

for $s = \sigma_{max}(A)$ and $u = \sigma_{min}(A)$, $\|B\|_2 = \sqrt{a - m^2}$, $\kappa_2(B) = \sqrt{\frac{a - u^2}{a - s^2}}$

where $a \geq s^2$

- for $a \to s^2$: $\kappa_2(B) \to +\infty$ and $\|B\|_2 \to \sqrt{s^2 - u^2}$
- for $a \to +\infty$: $\|B\|_2 \to +\infty$ and $\kappa_2(B) \to 1$

So there's a tradeoff, $\|B\|_2$ is increasing with a while $\kappa_2(B)$ decreases.

~~We will tackle two problems, minimize $\|B\|_2$ given $\kappa_2(B) \leq K$ and the more general case of minimizing $\|B\|_2$ and $\kappa_2(B)$ simultaneously without prescribing a bound to K and choosing the optimal a.~~

for the smallest $\|B\|_2$ with $\kappa_2(B) \le K$, $K>1$ fixed conditioned number.

$\frac{a-m^2}{a-s^2} \le K^2 \Rightarrow a-m^2 \le K^2(a-s^2) \Rightarrow a(K^2-1) \ge K^2 s^2 - m^2$

$\Rightarrow \quad a \ge \frac{K^2 s^2 - m^2}{K^2-1}$

$\|B\|_2$ increases with a, so $\|B\|_{2\,min} = \sqrt{\frac{K^2 s^2 - m^2}{K^2-1} - m^2} = \sqrt{\frac{K^2 s^2 - m^2 - K^2 m^2 + m^2}{K^2-1}}$

$\Rightarrow \|B\|_{2\,min} = K\sqrt{\frac{s^2-m^2}{K^2-1}}$, Overall, $\|B\|_2 \ge K\sqrt{\frac{s^2-m^2}{K^2-1}}$ with $\kappa_2(B) \le K$. □

- Prove that for every $A \in GL_n(\mathbb{R})$, the smallest possible spectral norm of a real matrix B, such that $A+iB$ is perfectly conditioned, is $\|B\|_{2\,min} = \sqrt{s^2-m^2}$ where $s = \sigma_{max}(A)$, $m = \sigma_{min}(A)$.

$\kappa_2(A+iB) = \frac{\sigma_{max}(C)}{\sigma_{min}(C)} = 1$ for $C = A+iB$ all its singular values must be equal therefore, $C^*C = \lambda I$ since the eigenvalues of C^*C are the squares of the singular values of C.

$C^* = A^T - iB^T \quad C^*C = (A^T - iB^T)(A+iB) = A^TA - iB^TA + iA^TB + B^TB = A^TA + B^TB + i(A^TB - B^TA)$

$\begin{cases} A^TA + B^TB = \lambda^2 I \\ A^TB - B^TA = 0 \end{cases}$ We will use only $A^TA + B^TB = \lambda I$ in our proof

$A^TA + B^TB = \lambda I \Rightarrow B^TB = \lambda I - A^TA$

but $x^TB^TBx = (Bx)^T(Bx) = \|Bx\|_2^2 \ge 0$ so B^TB is a positive semidefinite matrix and so its eigenvalues are nonnegative, and $\lambda I - A^TA$ is also positive semidefinite too $B^TB \ge 0 \Rightarrow \lambda I - A^TA \ge 0$

$\lambda_{max}(A^TA) = \sigma_{max}(A)^2 = s^2$, so the eigenvalues of $\lambda I - A^TA$ are:

$\lambda - \sigma_1(A)^2, \dots, \lambda - \sigma_n(A)^2$ and $\lambda - \sigma_i(A)^2 \ge 0 \Rightarrow \lambda \ge s^2$

for $B^TB = \lambda I - A^TA$, $\|B\|_2^2 = \lambda_{max}(B^TB)$, but the largest eigenvalue of $B^TB = \lambda I - A^TA$ is $\lambda - \lambda_{min}(A^TA) = \lambda - \sigma_{min}(A)^2 = \lambda - m^2$

$\Rightarrow \|B\|_2^2 = \lambda - m^2$ but $\lambda \geq s^2$ so $\lambda - m^2 \geq s^2 - m^2$

Finally, $\|B\|_2^2 \geq s^2 - m^2 \Rightarrow \|B\|_2 \geq \sqrt{s^2 - m^2}$

So our construction above is optimal because it achieves this and this is a global lower bound that applies to every real B for which $A + iB$ is perfectly conditioned, regardless of how B is constructed. □

- SVD construction and proof of $B_{QR} = B_{SVD}$.

The singular value decomposition of $A = U\Sigma V^T$, U and V are orthogonal and Σ is diagonal with the singular values of A, $\Sigma = diag(\sigma_1, \dots, \sigma_n)$, $\sigma_1 \geq \dots \geq \sigma_n \geq 0$

for $s = \sigma_1(A)$, $m = \sigma_n(A)$ and $D = \sqrt{s^2 I - \Sigma^2} = diag(\sqrt{s^2 - \sigma_1^2}, \dots, \sqrt{s^2 - \sigma_n^2})$

Choose $B = UDV^T$, $C = A + iB = U(\Sigma + iD)V^T$, $C^*C = V(\Sigma^2 + D^2)V^T = Vs^2 I V^T = s^2 I$

$\Rightarrow \kappa_2(C) = 1$ so C is perfectly conditioned

and $\|B\|_2 = \|UDV^T\|_2 = \|D\|_2 = \sqrt{s^2 - \sigma_n^2} = \sqrt{s^2 - m^2}$

for QR construction: $A = QR$, $M = s^2 R^{-T} R^{-1} - I$, $N = M^{1/2}$, $B_{QR} = QNR$

for SVD construction: $A = U\Sigma V^T$, $D = (s^2 I - \Sigma^2)^{1/2}$, $\Sigma = diag(\sigma_1, \dots, \sigma_n)$, $B_{SVD} = UDV^T$

$A = U\Sigma V^T$, $A = QR \Rightarrow R = Q^TA = Q^TV\Sigma V^T$, $W = Q^TV$: orthogonal $\Rightarrow R = W\Sigma V^T$

but $M = s^2 R^{-T} R^{-1} - I$ with $R^{-1} = V\Sigma^{-1}W^T$, $R^{-T} = W\Sigma^{-1}V^T$ so $R^{-T}R^{-1} = W\Sigma^{-2}W^T$

and $M = W(s^2\Sigma^{-2} - I)W^T$, $N = W(s^2\Sigma^{-2} - I)^{1/2}W^T$, so $B_{QR} = QW(s^2\Sigma^{-2} - I)^{1/2}W^TW\Sigma V^T$

$= QW(s^2\Sigma^{-2} - I)^{1/2}\Sigma V^T$ $QW = QQ^TU = U$ so $B_{QR} = U(s^2\Sigma^{-2} - I)^{1/2}\Sigma V^T$

and $(s^2\Sigma^{-2} - I)^{1/2} \cdot \Sigma = (s^2 I - \Sigma^2)^{1/2} = D$. Therefor, $B_{QR} = UDV^T = B_{SVD}$. □

①

- Theorem: Minimum-norm real imaginary part for perfect conditioning.

Let $A \in \mathbb{R}^{n \times n}$ be invertible, with largest and smallest singular values $s = \sigma_{max}(A)$, $m = \sigma_{min}(A)$

Then $\min_{\kappa_2(A+iB)=1} \|B\|_2 = \sqrt{s^2 - m^2}$ for $B \in \mathbb{R}^{n \times n}$. Moreover, this minimum is attained by i) SVD construction: if $A = U\Sigma V^T$ then $B_{SVD} = U(s^2 I - \Sigma^2)^{1/2}V^T$, and,

ii) QR construction: if $A = QR$, then $M = s^2 R^{-T}R^{-1} - I$, $N = M^{1/2}$ and $B_{QR} = QNR$.

Both constructions satisfy $B_{QR} = B_{SVD}$.

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Iterative Method for $Ax=b$

- Iterative method from theorem about perfect conditioning of $C=A+iB$.

We proved that for C to be perfectly conditioned we have $C^*C=\lambda I$ and for minimizing $\|B\|_2$, $C^*C=s^2I$ with $s=\sigma_{max}(A)$, $\mu=\sigma_{min}(A)$

Thus, as shown, $A^TA+B^TB=s^2I$, $Ax=b \Rightarrow A^TAx=A^Tb$

$\Rightarrow (s^2I-B^TB)x=A^Tb \Rightarrow s^2x-B^TBx=A^Tb \Rightarrow s^2x=A^Tb+B^TBx$

$\Rightarrow x_{k+1}=\frac{A^Tb}{s^2}+\frac{B^TB}{s^2}x_k$, $k\geq 0$

for $B^TB=s^2I-A^TA$ we have that $x_{k+1}=x_k-\frac{A^TA}{s^2}x_k+\frac{1}{s^2}A^Tb$

or just the two forms: (a) $x_{k+1}=\frac{1}{s^2}A^Tb+\frac{1}{s^2}B^TBx_k$, (b) $x_{k+1}=x_k+\frac{1}{s^2}A^T(b-Ax_k)$
that's Landweber iteration with optimal step $a=1/s^2$

Therefor, classical iterative gradient descent on $Ax=b$ is geometrically equivalent to extracting the real solution from its minimal perfectly conditioned complex counterpart.

Does it converge? Yes.

for $c=\frac{A^Tb}{s^2}$, $T=\frac{B^TB}{s^2}$, $x_{k+1}=Tx_k+c$, for x_*: exact solution
$x_*=c+Tx_*$: $x_{k+1}-x_*=T(x_k-x_*)$, error: $x_k-x_*=e_k$

$e_{k+1}=Te_k$, $e_1=Te_0$, $e_2=T^2e_0 \ldots$ $e_k=T^ke_0$

$T=\frac{B^TB}{s^2}=I-\frac{A^TA}{s^2}$ with $s=\sigma_1(A)\geq \ldots \geq \sigma_n(A)>0$
eigenvalues of T: $\lambda_i(T)=1-\frac{\sigma_i(A)^2}{s^2}$ so $\lambda_{max}(T)=1-\frac{\sigma_{min}(A)^2}{s^2}=1-\frac{m^2}{s^2}$

Since, $0<\sigma_i(A)\leq s \Rightarrow 0\leq\lambda_i(T)<1$, $\lambda_{max}(T)=\|T\|_2$ because T: symmetric positive semidefinite.

$\kappa_2(A) = \frac{s}{m}$ so $\|T\|_2 = 1 - \frac{1}{\kappa_2(A)^2} < 1$ since $\|T\|_2 < 1$

$e_k = T^k e_0 \Rightarrow \|e_k\|_2 = \|T^k e_0\|_2 \leq \|T^k\|_2 \cdot \|e_0\|_2$

$\Rightarrow \|x_k - x_*\|_2 \leq \left(1 - \frac{1}{\kappa_2(A)^2}\right)^k \|x_0 - x_*\|_2$ with $\lim_{k\to\infty} \left(1 - \frac{1}{\kappa_2(A)^2}\right)^k = 0$

and therefore, $\lim_{k\to\infty} x_k = x_*$ so the sequence (x_k) converges. □

Theorem 2: Fixed-point induced by the minimum-norm construction.

Let $A \in GL_n(\mathbb{R})$, with singular values $s = \sigma_{max}(A)$, $m = \sigma_{min}(A) > 0$.
Let $B \in \mathbb{R}^{n\times n}$ be the matrix attaining the minimum norm from Theorem 1. with $\|B\|_2 = \sqrt{s^2 - m^2}$, $\kappa_2(A+iB) = 1$. Then $B^TB = s^2 I - A^TA$.
For any $b \in \mathbb{R}^n$ consider $x_{k+1} = \frac{A^Tb}{s^2} + \frac{B^TB}{s^2} x_k$, $k \geq 0$.
Then the following statements hold:
i) Convergence: for initial vector $x_0 \in \mathbb{R}^n$, the sequence (x_k) converges to the unique solution $x_* = A^{-1}b$.
ii) convergence rate: $\forall k \geq 0$, $\|x_k - x_*\|_2 \leq \left(1 - \frac{m^2}{s^2}\right)^k \|x_0 - x_*\|_2$
equivalently, the contraction factor is: $q = \left(1 - \frac{1}{\kappa_2(A)^2}\right) < 1$.

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Generalized Theorem 1

· Theorem 1 but for $\kappa_2(C) \leq K$, with $K \geq 1$.

$\min_{B \in \mathbb{R}^{m \times n}, \kappa_2(A+iB) \leq K} \|B\|_2$, for $K \geq \kappa_2(A)$ set $B = 0$

for $1 \leq K \leq \kappa_2(A)$, B must be nonzero, set $t = \|B\|_2$.
with v_s the unit right singular vector of A corresponding to s and v_m corresponding to m, $\|v_s\| = 1$ and $\|Av_s\| = s$.
Because, A, B, and v_s are real: $\|Cv_s\|_2^2 = \|(A+iB)v_s\|_2^2 = \|Av_s\|_2^2 + \|Bv_s\|_2^2 = s^2 + \|Bv_s\|_2^2 \geq s^2$
Therefor, $\sigma_{max}(C) \geq \|Cv_s\|_2^2 \Rightarrow \sigma_{max}(C) \geq s^2$.
Adding iB cannot reduce the largest singular value below that of A.
for $\|v_m\| = 1$ and $\|Av_m\|_2 = m$, $\|Cv_m\|_2^2 = \|Av_m\|_2^2 + \|Bv_m\|_2^2 = m^2 + \|Bv_m\|_2^2 \leq m^2 + t^2$
because $\|Bv_m\|_2 \leq \|B\|_2 = t$ and so $\sigma_{min}(C) \leq \|Cv_m\|_2 \Rightarrow \sigma_{min}(C)^2 \leq m^2 + t^2$

$\kappa_2(C)^2 = \frac{\sigma_{max}(C)^2}{\sigma_{min}(C)^2} \geq \frac{s^2}{m^2+t^2}$ where $t = \|B\|_2$ but $\kappa_2(C) \leq K$ so $K^2 \geq \frac{s^2}{m^2+t^2}$

$\Rightarrow t^2 \geq \frac{s^2}{K^2} - m^2 \Rightarrow \|B\|_2^2 \geq \frac{s^2}{K^2} - m^2$ and $\|B\|_2^2 \geq 0$, for $\kappa_2(A+iB) \leq K$.

To sum up, global lower bound $\forall B \in \mathbb{R}^{m \times n}$: $\|B\|_2 \geq \sqrt{\max\{0, \frac{s^2}{K^2} - m^2\}}$. □

· Calculate the matrix that attains the global lower bound with the SVD and QR constructions from theorem 1.

(a) SVD: $A = U\Sigma V^T$, Σ diagonal and V and U orthogonal
set $\tau = \frac{s}{K}$, raise every $\sigma_i < \tau$ to τ while leaving others unchanged.
$d_i = \sqrt{\max\{0, \tau^2 - \sigma_i^2\}}$, $D = \mathrm{diag}(d_1, \dots, d_n)$
$D^2 = (\tau^2 I - \Sigma^2)_+$ where L_+ is the positive part of a real symmetric matrix L obtained by replacing its negative eigenvalues with zero.

if $\sigma_i < \tau$: $d_i = \sqrt{\tau^2 - \sigma_i^2}$, else $d_i = 0$, choose $B_* = UDV^T$
$C = A + iB = U(\Sigma + iD)V^T$, but U and V^T do not change the singular values of $\Sigma + iD$

$(\Sigma+iD)^*(\Sigma+iD) = \Sigma^T\Sigma + D^TD + i\Sigma^TD - iD^T\Sigma$ with $\Sigma^T=\Sigma$, $D^T=D$ as diagonal

So $(\Sigma+iD)^*(\Sigma+iD) = \Sigma^2+D^2$

the ith entry of Σ^2+D^2 is $\sigma_i^2+d_i^2 = \sigma_i^2+\max\{0, \tau^2-\sigma_i^2\} = \max\{\sigma_i^2, \tau^2\}$.

So $\sigma_i(C) = \max\{\sigma_i(A), \tau\}$ with $\tau = \frac{s}{k}$, $\sigma_{max}(C)=\sigma_{max}(A)=s$, $\sigma_{min}(C)=\max\{m,\tau\}$

Therefor, $\kappa_2(C) = \frac{s}{\max\{m, s/k\}} \le k$

$\|B_x\|_2 = \|D\|_2 = \max d_i$ but $d_i^2 = \max\{0, \tau^2-\sigma_i^2\}$, $\max d_i = \sqrt{\max\{0, \frac{s^2}{k^2}-m^2\}}$ → global lower bound

(b) QR: $A=QR$, $\tau=\frac{s}{k}$ define $P=(\tau^2 I - R^TR)_+$, $N=(R^{-T}PR^{-1})^{1/2}$

$B_{QR} = QNR = Q(R^{-T}PR^{-1})^{1/2}R$, P is positive semidefinite so $R^{-T}PR \ge 0$

and so $(R^{-T}PR^{-1})^{1/2}$ exists

$B_{QR}^T B_{QR} = (QNR)^T(QNR) = QN^2R = R^TR^{-T}PR^{-1}R = P = (\tau^2 I - R^TR)_+$

but $C^*C = A^TA + B_{QR}^TB_{QR} = R^TR + (\tau^2 I - R^TR)_+$

~~so the eigenvalues of C are~~ The singular values of C: $\sigma_i(A)^2 + \max\{0, \tau^2-\sigma_i(A)^2\}$

$\Rightarrow \sigma_i(C) = \max\{\sigma_i(A), \tau\}$ with $\sigma_{max}(C)=s$, $\sigma_{min}(C)=\max\{m,\tau\}$

$B_{QR}^TB_{QR} = P$ so $\|B_{QR}\|_2^2 = \lambda_{max}(P) = \max\{0, \tau^2-m^2\}$

$\Rightarrow \|B_{QR}\|_2 = \sqrt{\max\{0, s^2/k^2-m^2\}}$. □

Theorem 3: Minimum-norm perturbation under a prescribed condition number bound.

Let $A \in GL(\mathbb{R})$, with $s=\sigma_{max}(A)$, $m=\sigma_{min}(A)>0$, for any $k\ge 1$ consider the optimization problem with $\kappa_2(A+B) \le k$ $\|B\|_2 \to min$, then $\|B\|_{2,min} = \sqrt{\max\{0, s^2/k^2-m^2\}}$.

Moreover, let $A=U\Sigma V^T$ the singular value decomposition of A.

Define $D = (\frac{s^2}{k^2}I - \Sigma^2)_+^{1/2}$, the matrix $B_x = UDV^T$ attains that minimum

Same for $B_x = Q(R^{-T}PR^{-1})^{1/2}R$ where $A=QR$ the QR-decomposition of A and $P=(\tau^2 I - R^TR)_+$.

In any case, $\kappa_2(A+B) = \frac{s}{\max\{m, s/k\}} \le k$.